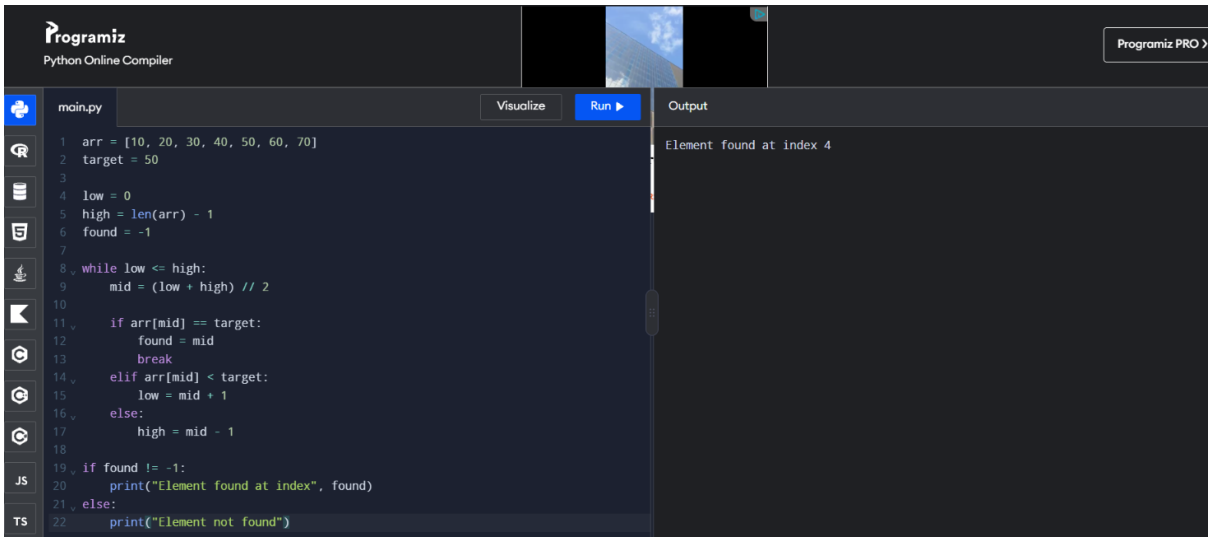
4. Binary Search with Big-O Analysis

Aim

To search for a given element in a sorted array using the Binary Search algorithm and analyze its time complexity.

Procedure

1. Read the sorted array and the target element.
2. Initialize low and high indices.
3. Find the middle element of the array.
4. Compare the target element with the middle element.
5. If they are equal, return the position.
6. If the target is greater, search the right half.
7. If the target is smaller, search the left half.
8. Repeat until the element is found or the search space becomes empty.
9. Display the result.



Result

Thus, the given element was searched successfully using the Binary Search algorithm.